Alex Dytso

CONTACT INFORMATION

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CURRENT POSITION

Princeton University Electrical Engineering Department

Postdoctoral Research Associate — September 2016 - Currently
With Prof. H. Vincent Poor

EDUCATION

University of Illinois at Chicago

Ph.D., Electrical and Computer Engineering, May 2016
- Dissertation Title: *Discrete Inputs in Gaussian Interference Channels: Performance Analysis and Approximate Optimality*
- Advisors: Prof. Daniela Tuninetti & Prof. Natasha Devroye

University of Illinois at Chicago

M.Sc., Electrical Engineering, December 2014

University of Illinois at Chicago

B.Sc., Electrical and Computer Engineering, May 2011

RESEARCH INTERESTS & AREAS OF EXPERTIES

- *Interference Mitigation in Wireless Networks:* Study of optimal transmission schemes for interference channels with special emphasis on using discrete input constellations (**[J.9]**, **[J.7]**, **[C.25]**, **[C.26]**, **[C.27]**, **[C.28]**, **[C.32]**).
- *Information and Estimation Relations:* Study of relations between information and estimation measures with the focus on using such identities (or inequalities) in characterizing information theoretic capacity (**[J.8]**, **[C.12]**, **[C.14]**, **[C.21]**, **[C.22]**, **[C.23]**).
- *Differential Privacy:* Rigorous study of privacy guarantees (**[C.16]**,**[J.19]**).
- *Noisy Order Statistics:* Estimation of permuted noisy data (**[J.16]**, **[C.7]**).
- *Limits of Statistical Signal Recovery:* Study of fundamental lower bounds on the precision of a random signal recovery (**[J.6]**, **[C.5]**, **[J.15]**, **[C.24]**).
- *Convex Optimization Over Metric Spaces:* Optimization of convex operators over metric spaces with a special emphasis on the probability space (**[C.13]**, **[C.17]**, **[C.18]**, **[J.14]**, **[J.15]**, **[J.20]**).
- *Estimation in Poisson Noise with Applications to Laser Communications:* Study of optimal Bayesian estimators in Poisson noise (**[J.13]**, **[C.1]**, **[C.3]**).
- *Cognitive Networks:* Study of ultimate performance of cognitive networks with the emphasis on interaction of a cognitive user with the legacy equipment (**[J.10]**, **[J.11]**, **[C.30]**, **[C.31]**, **[C.33]**).
- *Capacity of Channels with Amplitude Constraints:* Study of the capacity of additive channels with an amplitude constraint on the input. Amplitude constrained channels commonly occur in practice, however, neither capacity nor optimal transmission schemes are known for such channels in general (**[J.2]**, **[J.3]**, **[J.14]**, **[J.17]**, **[C.12]**, **[C.20]**, **[C.21]**).

- *Capacity of Non-Gaussian Noise Channels:* Study of the capacity of non-Gaussian additive channels with the specific focus on impulsive noise channels that can be modeled with generalized Gaussian distributions (**[J.4]**, **[J.17]**, **[C.10]**, **[C.21]**, **[C.22]**).
- *Entropy Power Inequalities:* Study of entropy power inequalities with the emphasis on finding condition for the equality and stability (**[J.18]**, **[C.19]**).

Professional Affiliations

Vocational

- **Lecturer**, *Princeton University Operations Research and Financial Engineering (ORF)*, September 2018- January 2019.
  ORF 245: Fundamentals of Statistics. Note 1: Instructor. Course size 150 students. Note 2: Received the best course evaluation over the 10 year period.
- ◇ **Internship**, *NASA, Goddard Space Flight Center*, Maryland, USA, Summer 2014.
  Error Correction Codes for TDRSS satellites.
- ◇ **Internship**, *China Solar and Clean Energy Solutions Inc.*, Beijing, China, Summer 2011.
  Assistant to the main engineer. Performed efficiency of deployment calculations on solar water heaters and solar panels. Developed International Confidence, International Sensitivity and Awareness.
- ◇ **Math Tutor**, *Mathematics, Statistics, and Computer Science Tutoring Center*, UIC, Chicago, USA, 2008-2011
  Assisted students with understanding and solving low and high-level math problems. Formed study groups.

Miscellaneous

- ◇ **Repairing Technician**, *DRO Electric Co.*, Chicago.
  Setting up, installing and repairing air conditioning units.
- ◇ **Installation Technician**, *SH Electric Inc.*, Chicago.
  Installation security cameras, speaker systems, fire alarms, light fixtures, electrical wiring etc.

Programming Skills

MatLab, C++, Java

Programming Projects

- ◇ **Tight MMSE Bounds for the AGN Channel Under KL Divergence Constraints on the Input Distribution (2018)**. This code was designed to supplement **[J.15]**. The code is available at https://codeocean.com/capsule/4970811/tree/v1.
- ◇ **Optimal Higher-Order Constellation Design and Least-Favorable Distributions (2018)** This code was designed to supplement **[J.3]**. The code is available at https://doi.org/10.5281/zenodo.1401556.
- ◇ **MATLAB Projects (2011-Current)** Full list of MATLAB projects can be viewed on MatLabCentral.com. Projects include: Channel Equalizers, scalar Kalman Filter, $3-D$ Extended Kalman Filter, Combinatorial functions.
- ◇ **Beowulf computer cluster (2012)** Using high computing capabilities of computer clusters to simulate and approximate Network Information problems.

- ◇ **Senior Design Project "Watch & White Board" (2009)** Using Chronos eZ-430 watch to translate white board notes into the digital format.
- ◇ **Micro-Controller Project (2010)** All designs were performed on the Motorola 68HC11 micro-controller. Projects Included: Stop-Watch/Clock, Count-Down/Up timers, Four-Function Calculator, Electronic Flashcards, Voltmeter.

Grant Writing Experience

- ◇ Co-authored the proposal for NSF-CIF-BSF (PI: H. Vincent Poor, Co-PI: Shlomo Shamai). The proposal was funded.

Talks & Seminars

- ◇ "Some Aspects of Totally Positive Kernels Useful in Information Theory," The 1st International Workshop on Mathematical Tools and technologies for IoT and mMTC Networks Modeling, Marocco, April 2019.
- ◇ "Generalized Gaussian Distributions in Estimation and Information Theory," *Rutgers University*, Host: Roy Yates, October 2018.
- ◇ "A View of Information-Estimation Relations in Gaussian Networks," *Technion*, Host: Shlomo Shamai, May 2018.
- ◇ "On the Structure of the Least Favorable Distributions," *New Jersey Institute of Technology (NJIT)*, Host: Joerg Kliewer, March 2018.
- ◇ "A View of Information-Estimation Relations in Gaussian Networks," *Princeton University*, Host: Paul Cuff, February 2017.
- ◇ "On the Minimum Mean p-th Error in Gaussian Noise Channels and its Applications," *Bell Labs Nokia*, Host: Henry Landau, October 2016.
- ◇ "On the Minimum Mean p-th Error in Gaussian Noise Channels and its Applications," *New Jersey Institute of Technology (NJIT)*, Host: Osvaldo Simeone, October 2016.
- ◇ "On the Minimum Mean p-th Error in Gaussian Noise Channels and its Applications," *Princeton University*, Host: H. Vincent Poor, September 2016.
- ◇ "Overview and New Advances on the Interplay Between Estimation and Information Measures," *Illinois Institute of Technology (IIT)*, Host: Salim El Rouayheb, September 2016.
- ◇ "On the Optimality of Treating Interference as Noise in Competitive Scenarios," *University of Maryland College Park*, Host: Sennur Ulukus, July 2016.

Teaching & Mentoring Experience

Teaching

Princeton University

- ◇ ORF 245: Fundamentals of Statistics. Note 1: Instructor. Course size 150 students. Note 2: Received the best course evaluation over the 10 year period.

University of Illinois at Chicago

- ◇ ECE 437: Wireless Communications. Topic: Random Processes, Fall 2014, Spring 2014. Note: Part time instructor.
- ◇ ECE 531: Information Theory. Topic: Capacity of the Continuous Time Channels, Parallel Channels and Water Filling, Fall 2015. Note: Part time instructor.

Mentoring

- *Research Supervision.* I am currently a mentor for Mert Al who is a Ph.D. student in the department of Electrical Engineering at Princeton University. Our collaboration has result in one journal submission **[J.3]**.
- *Research Supervision.* I am currently a mentor for Semih Yagli who is a Ph.D. student in the department of Electrical Engineering at Princeton University. Our collaboration has result in one journal submission **[J.14]** and one conference paper **[C.9]**.

◇ *Research Supervision.* I am currently a mentor for Wei Cao who is a Ph.D. student at the University of Electronic Science and Technology of China. Wei Cao is currently a visiting student in the department of Electrical Engineering at Princeton University. Wei Cao work focuses on massive access techniques for wireless networks. Our collaboration has resulted in one journal submission **[J.5]** and two conference submissions **[C.8]**, **[C.11]**.

◇ *Research Supervision.* I am currently a mentor for Thee Chanyaswad who is currently a Ph.D. student in the department of Electrical Engineering at Princeton University. Our joint collaboration resulted in submissions of conference **[C.16]** and journal paper **[J.19]** on the differential privacy.

◇ *Research Supervision.* I have been a mentor for Naveed Naimipour as a part of his undergraduate research study at UIC. Naveed has recently joint a research group of Daniela Tuninetti and Natasha Devroye as a Ph.D. student.

Tutoring

◇ *Mathematics, Statistics, and Computer Science Tutoring Center* at University of Illinois at Chicago. *Math Tutor.* 2007-2011.

◇ *Free Community Tutoring Center* at St. Volodymyr Church, Chicago. *Organizer, Tutor.* 2011-2012.

Honors & Awards

◇ **Wexler Award of Excellence.** Awarded to promising graduate students. 2011-2012

◇ **International Engineering Consortium's William L. Everitt Student Award of Excellence.** Honors outstanding seniors who have demonstrated an interest in the communications field. 2011

◇ **UIC Engineering Alumni Association Scholarship.** Awarded each year to two students who demonstrated high academic achievements and strong desire to enhance their education by demonstrating leadership skills. 2010

◇ **Irene Jones Scholarship.** Support the next generation of engineers - the men and women who will change the world. 2010

Professional Service

Technical Program Chair / Conference Organization

- Workshop on Physical-Layer Methods for Security and Privacy in 5G and the IoT at IEEE CNS '19, Washington D.C., USA, June 2019.
- IEEE International Conference on Computing, Networking and Communications (ICNC 2019), Honolulu, USA, February 2019.
- The 4th Workshop on Physical-Layer Methods for Wireless Security (PLMWS) at IEEE CNS '17, Las Vegas, USA, October 2017.

Technical Program Committees

- 3nd International Workshop on Non-Orthogonal Multiple Access Techniques for 5G at IEEE GLOBECOM 2018 in Abu Dhabi, UAE, December 2018.
- 2nd International Workshop on Non-Orthogonal Multiple Access Techniques for 5G at IEEE GLOBECOM 2017 in Singapore, December 2017.

- 52th Annual Conference on Information Sciences and Systems (CISS) in Princeton, USA, March 2018.
- ICC 2018 Workshop - NOMA for 5G in Kansas City, USA, May 2018.

Editorial

- Guest-Editor for the special issue on "Wireless Networks: An Information Theoretic Perspective" in Entropy journal.

Organized Sessions

- "Simulates Communication and Energy Transfer" session at the Conference on Information Sciences and Systems (CISS), Princeton, USA, March 2019. Organized with Bruno Clerckx.
- "On the Limits of Recovering Permuted Data" session at the Conference on Information Sciences and Systems (CISS), Princeton, USA, March 2019. Organized with Martina Cardone.
- "Optimal Communication with Discrete Input Signals" session at the Conference on Information Sciences and Systems (CISS), Princeton, USA, March 2018. Organized with M. Goldenbaum, H. V. Poor, and S. Shamai.

Reviewer

- Elsevier Signal Processing
- IEEE Transactions on Information Theory
- IEEE Transactions on Wireless Communications
- IEEE Transactions on Communications
- IEEE Access
- IEEE Journal on Selected Areas in Communications
- IEEE Globecom
- IEEE ICC
- IEEE International Symposium on Information Theory (ISIT)
- IEEE Information Theory Workshop (ITW)
- Indian Journal of Mathematics (IJM)

Public Services and Volunteering

- Information Theory Society Chicago Chapter, Chicago. *One of the chapter organizers. Prepared the paperwork, helped to collect petition signatures.*
- Future City Competition, Chicago. *Judge from local IEEE section.*
- Student Information Theory Society. *Student Member. Organized student events:*
  - ISIT 2016: Session Chair Assistant.
  - ISIT 2015: Session Chair Assistant.
  - ISIT 2014: Helped organized meeting between Shannon awardee and students.
  - Allerton 2012: Organized student events."Pictionary".
- Free Community Tutoring Center. St. Volodymyr Church, Chicago. *Organizer, Tutor.*
- UIC Flourish of Chicago (an open source conference), Chicago. *Volunteer.*

Memberships

IEEE, Information Theory Society

Language Skills

Fluent in Russian, Fluent in Ukrainian, Basic Polish

PUBLICATIONS

Journal Papers

Published

[J.1] **A. Dytso**, M. Cardone, H. V. Poor, "On Estimating the Norm of a Gaussian Vector under Additive White Gaussian Noise," to appear.

[J.2] **A. Dytso**, M. Goldenbaum, H. V. Poor, S. Shamai (Shitz), "Amplitude Constrained MIMO Channels: Properties of Optimal Input Distributions and Bounds on the Capacity," *Entropy*, Vol. 21, No. 2, February 2019 .

[J.3] **A. Dytso**, H. V. Poor, S. Shamai (Shitz), "On the Capacity of the Peak Power Constrained Vector Gaussian Channel: An Estimation Theoretic Perspective," arXiv:1804.08524 [cs.IT] (to appear in IEEE Transactions on Information Theory).

[J.4] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "Analytical Properties of Generalized Gaussian Distributions," *Journal of Statistical Distributions and Applications*, 5:6, 2018.

[J.5] W. Cao, **A. Dytso**, G. Feng, H. V. Poor, Z. Chen, "Differentiated Service-Aware Group-Paging for Massive Machine-Type Communications," *IEEE Transactions on Communications*, Vol. 66, No. 11, November 2018.

[J.6] **A. Dytso**, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, S. Shamai (Shitz), "On the Minimum Mean p-th Error in Gaussian Noise Channels and its Applications," *IEEE Transactions on Information Theory*, Vol. 64, No. 3, March 2018.

[J.7] **A. Dytso**, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, S. Shamai (Shitz), "On Communication through a Gaussian Channel with an MMSE Disturbance Constraint," *IEEE Transactions on Information Theory*, Vol. 64, No. 1, January 2018.

[J.8] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "A View of Information-Estimation Relations in Gaussian Networks," *Entropy*, Vol. 19, No. 8, August 2017.

[J.9] **A. Dytso**, D. Tuninetti and N. Devroye, "Interference as Noise: Friend or Foe?," *IEEE Transactions on Information Theory*, Vol. 62, No. 6, June 2016.

[J.10] **A. Dytso**, D. Tuninetti and N. Devroye, "On the Two-User Interference Channel with Lack of Knowledge of the Interference Codebook at One Receiver," *IEEE Transactions on Information Theory*, Vol. 61, No. 3, March 2015.

[J.11] **A. Dytso**, D. Tuninetti and N. Devroye, "On the Capacity Region of the Two-User Interference Channel with a Cognitive Relay," *IEEE Transactions on Wireless Communications*, Vol. 13, No. 12, December 2014.

Preprints

[J.12] A. Yin, **A. Dytso**, D. Ramirez, H. V. Poor, "Agressive Fraud Prevention via Reinforcement Learning" (in preparation).

[J.13] **A. Dytso**, H. V. Poor, "Estimation in Poisson Noise: Properties of The Conditional Mean Estimator," (submitted to IEEE Transactions on Signal Processing)

[J.14] **A. Dytso**, S. Yagli, H. V. Poor, and S. Shamai (Shitz),"The Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel: An Upper Bound on the Number of Mass Points," (Submitted to IEEE Transactions on Information Theory) arXiv:1901.03264

[J.15] **A. Dytso**, M. Fauß, A. M. Zoubir, H. V. Poor, "Tight MMSE Bounds for the AGN Channel Under KL Divergence Constraints on the Input Distribution," (submitted to IEEE Transactions on Signal Processing)

[J.16] **A. Dytso**, M. Cardone, M. S. Veedu, H. V. Poor, "On Estimation under Noisy Order Statistics," arXiv:1901.06294

[J.17] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "Communication over Channels with Generalized Gaussian Noise," (in preparation).

[J.18] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "On the Equality Condition for the I-MMSE Proof of the Entropy Power Inequality," arXiv:1703.07442 (in preparation).

[J.19] T. Chanyaswad, **A. Dytso**, P. Mittal, and H. V. Poor, "MVG Mechanism: Differential Privacy Under Matrix-Valued Query," arXiv: 1801.00823.

[J.20] **A. Dytso**, M. Goldenbaum, H. V. Poor, S. Shamai (Shitz), "When Are Discrete Alphabets Optimal? - Optimization Methods and New Results," (in preparation).

Conference Papers

2019

[C.1] **A. Dytso** and H. V. Poor, "On Stability of Linear Estimators in Poisson Noise," *Asilomar Conference on Signal, Systems, and Control*, Pacific Grove, CA, USA, submitted.

[C.2] W. Cao, **A. Dytso**, M. Fauß, G. Feng, and H. V. Poor, "Robust Power Allocation for Parallel Gaussian Channels with Approximately Gaussian Input Distributions," *IEEE Global Communications Conference (GLOBECOM)*, Hawaii, USA, December 2019, submitted.

[C.3] **A. Dytso** and H. V. Poor, "Properties of The Conditional Mean Estimator in Poisson Noise," *IEEE Information Theory Workshop (ITW)*, Visby, Gotland, Sweden, August 2019, accepted.

[C.4] S. Yagli, **A. Dytso**, H. V. Poor, "Estimation of Bounded Normal Mean: An Alternative Proof for the Discreteness of the Least Favorable Prior," *IEEE Information Theory Workshop (ITW)*, Visby, Gotland, Sweden, August 2019, accepted.

[C.5] Michael Fauß, Abdelhak M. Zoubir, **Alex Dytso**, H. Vincent Poor, Nagananda K. G, "Tight Bounds on the Weighted Sum of MMSEs with Applications in Distributed Estimation," *IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, Cannes, France, July 2019, to appear.

[C.6] S. Yagli, **A. Dytso**, H. V. Poor, S. Shamai (Shitz), "An Upper Bound on the Number of Mass Points in the Capacity Achieving Distribution for the Amplitude Constrained Additive Gaussian Channel," *IEEE International Symposium on Information Theory (ISIT)*, Paris, France, June 2019.

[C.7] **A. Dytso**, M. Cardone, M. S. Veedu, H. V. Poor, "On Estimation under Noisy Order Statistics," *IEEE International Symposium on Information Theory (ISIT)*, Paris, France, June 2019.

[C.8] W. Cao, **A. Dytso**, Y. Shkel, G. Feng, H. V. Poor, "Sum-Capacity of the MIMO Gaussian Many-Access Channel," *IEEE International Conference on Communications (ICC)*, Shanghai, China, May 2019.

[C.9] S. Yagli, **A. Dytso**, H. V. Poor, S. Shamai (Shitz), "Some Aspects of Totally Positive Kernels Useful in Information Theory," *IEEE Wireless Communications and Networking Conference (WCNC)*, Marrakech, Morocco, April 2019, invited.

2018

[C.10] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "On Lossy Compression of Generalized Gaussian Sources," *IEEE International Conference on the Science of Electrical Engineering (ICSEE)*, Eilat, Israel, December 2018.

[C.11] W. Cao, **A. Dytso**, G. Feng, H. V. Poor, Z. Chen, "Group Paging for Massive Machine-Type Communications with Diverse Access Requirements," *IEEE Global Communications Conference (GLOBECOM)*, Abu Dhabi, UAE, December 2018.

[C.12] **A. Dytso**, M. Egan, S. M. Perlaza, H. V. Poor, S. Shamai (Shitz), "Capacity of the Vector Gaussian Channel in the Small Amplitude Regime," *IEEE Information Theory Workshop (ITW)*, Guangzhou, China, November 2018.

[C.13] **A. Dytso**, H. V. Poor, S. Shamai (Shitz), "Optimal Inputs for Some Classes of Degraded Wiretap Channels," *IEEE Information Theory Workshop (ITW)*, Guangzhou, China, November 2018.

[C.14] M. Fauß, **A. Dytso**, A. M. Zoubir, H. V. Poor, "Tight MMSE Bounds for the AGN Channel Under KL Divergence Constraints on the Input Distribution," *IEEE Statistical Signal Processing Workshop (SSP)*, Freiburg, Germany, June 2018.

[C.15] T. Chanyaswad, **A. Dytso**, P. Mittal, and H. V. Poor, "MVG Mechanism: Differential Privacy Under Matrix-Valued Query," *Proceedings of the 25th ACM Conference on Computer and Communications Security*, Toronto, Canada, October 2018, to appear.

[C.16] G. Reeves, H. D. Pfister, **A. Dytso**, "Mutual Information as a Function of Matrix SNR for Linear Gaussian Channels," *IEEE International Symposium on Information Theory (ISIT)*, Vail, USA, June 2018.

[C.17] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "On the Structure of the Least Favorable Prior Distributions," *IEEE International Symposium on Information Theory (ISIT)*, Vail, USA, June 2018.

[C.18] **A. Dytso**, M. Goldenbaum, H. V. Poor, and S. Shamai(Shitz), "When Are Discrete Channel Inputs Optimal? - Optimization Methods and New Results," *52th Annual Conference on Information Sciences and Systems (CISS)*, Princeton, USA, March 2017.

2017

[C.19] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "On the Equality Condition for the I-MMSE Proof of the Entropy Power Inequality," *55th Annual Allerton Conference on in Communication, Control, and Computing (Allerton)*, Monticello, USA, October 2017.

[C.20] **A. Dytso**, M. Goldenbaum, H. V. Poor, S. Shamai (Shitz), "Upper and Lower Bounds on the Capacity of Amplitude-Constrained MIMO Channels," *IEEE Global Communications Conference (GLOBECOM)*, Singapore, December 2017.

[C.21] **A. Dytso**, R. Bustin, H. V. Poor, S. Shamai (Shitz), "On Additive Channels with Generalized Gaussian Noise," *IEEE International Symposium on Information Theory (ISIT)*, Aachen, Germany, June 2017.

[C.22] **A. Dytso**, M. Goldenbaum, H. V. Poor, S. Shamai (Shitz), "A Generalized Ozarow-Wyner Capacity Bound with Applications," *IEEE International Symposium on Information Theory (ISIT)*, Aachen, Germany, June 2017.

2016

[C.23] **A. Dytso**, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, S. Shamai (Shitz), "On the Applications of the Minimum Mean p-th Error (MMPE) to Information Theoretic Quantities," *IEEE Information Theory Workshop (ITW)*, Cambridge, UK, September 2016.

[C.24] **A. Dytso**, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, S. Shamai (Shitz), "On the Minimum Mean p-th Error in Gaussian Noise Channels and its Applications," *IEEE International Symposium on Information Theory (ISIT)*, Barcelona, Spain, June 2016.

[C.25] **A. Dytso**, R. Bustin, D. Tuninetti, N. Devroye, H. V. Poor, S. Shamai (Shitz), "On Communications Through a Gaussian Noise Channel with an MMSE Disturbance Constraint," *IEEE International Symposium on Information and its Applications (ITA)*, San Diego, USA, February 2016.

2015

[C.26] **A. Dytso**, N. Devroye and D. Tuninetti, "Nearly Optimal non-Gaussian Codes for the Gaussian Interference Channel," *49th Asilomar Conference on Signals, Systems and Computers*, Asilomar, USA October 2015.

[C.27] **A. Dytso**, N. Devroye and D. Tuninetti, "i.i.d. Mixed Inputs and Treating Interference as Noise are gDoF Optimal for the Symmetric Gaussian Two-user Interference Channel," *IEEE International Symposium on Information Theory (ISIT)*, Hong Kong, June 2015.

[C.28] **A. Dytso**, D. Tuninetti and N. Devroye, "On the Two-User Interference Channel with Partial Codebook Knowledge at One Receiver: Symmetric Capacity to with a Gap with PAM Inputs," *IEEE Information Theory Workshop (ITW)*, Jerusalem, Israel, April 2015.

2014

[C.29] **A. Dytso**, N. Devroye and D. Tuninetti, "On Gaussian Interference Channels with Mixed Gaussian and Discrete Inputs," *IEEE International Symposium on Information Theory (ISIT)*, Honolulu, USA, June 2014.

[C.30] **A. Dytso**, D. Tuninetti and N. Devroye, "On Discrete Alphabets for the Two-User Gaussian Interference Channel with One Receiver Lacking Knowledge of the Interfering Codebook," *Information Theory and Applications Workshop (ITA)*, Pacific Grove, USA, February 2014.

2013

[C.31] **A. Dytso**, N. Devroye and D. Tuninetti, "On the Capacity of Interference Channels with Partial Codebook Knowledge," *IEEE International Symposium on Information Theory (ISIT)*, Istanbul, Turkey, July 2013.

2012

[C.32] **A. Dytso**, N. Devroye and D. Tuninetti, "The Sum-Capacity of the Symmetric Linear Deterministic Complete K-user Z-Interference Channel," *50th Annual Allerton Conference on in Communication, Control, and Computing (Allerton)*, Monticello, USA, October 2012.

[C.33] **A. Dytso**, N. Devroye and D. Tuninetti, "On The Capacity of the Symmetric Interference Channel with a Cognitive Relay at High SNR," *IEEE International Conference on Communications (ICC)*, Ottawa, Canada, June 2012.